

3. Design of a moving average filter using Verilog

Aim

To design and simulate a Moving Average Filter using RTL Verilog to smooth input data by averaging a fixed number of recent samples.

Software Required

- Xilinx Vivado Design Suite
- Vivado Simulator
- PC or Laptop

RTL for Moving average filter

```
module moving_average_filter (  
    input wire    clk,  
    input wire    rst,  
    input wire [7:0] data_in,  
    output reg [7:0] data_out  
);  
reg [7:0] shift_reg [0:3]; // Store last 4 samples  
reg [9:0] sum;           // Sum of 4 samples (max = 1020)  
integer i;  
always @(posedge clk or posedge rst) begin  
    if (rst) begin  
        for (i = 0; i < 4; i = i + 1)  
            shift_reg[i] <= 0;  
        data_out <= 0;  
        sum <= 0;  
    end else begin  
        // Shift samples  
        shift_reg[3] <= shift_reg[2];  
        shift_reg[2] <= shift_reg[1];  
        shift_reg[1] <= shift_reg[0];  
        shift_reg[0] <= data_in;
```

```

        sum <= shift_reg[0] + shift_reg[1] +
            shift_reg[2] + shift_reg[3];
        data_out <= sum >> 2;
    end
end
endmodule

```

Testbench for Moving average filter

```

`timescale 1ns / 1ps

module tb_moving_average_with_noise;

    reg clk;

    reg rst;

    reg [7:0] data_in;
    wire [7:0] data_out;

    real sine_val;

    integer noise;

    integer temp;

    integer i;

    moving_average_filter dut (
        .clk(clk),
        .rst(rst),
        .data_in(data_in),
        .data_out(data_out)
    );

    always #5 clk = ~clk;

    initial begin

        clk = 0;

        rst = 1;

        data_in = 0;

        #20 rst = 0;
    end
endmodule

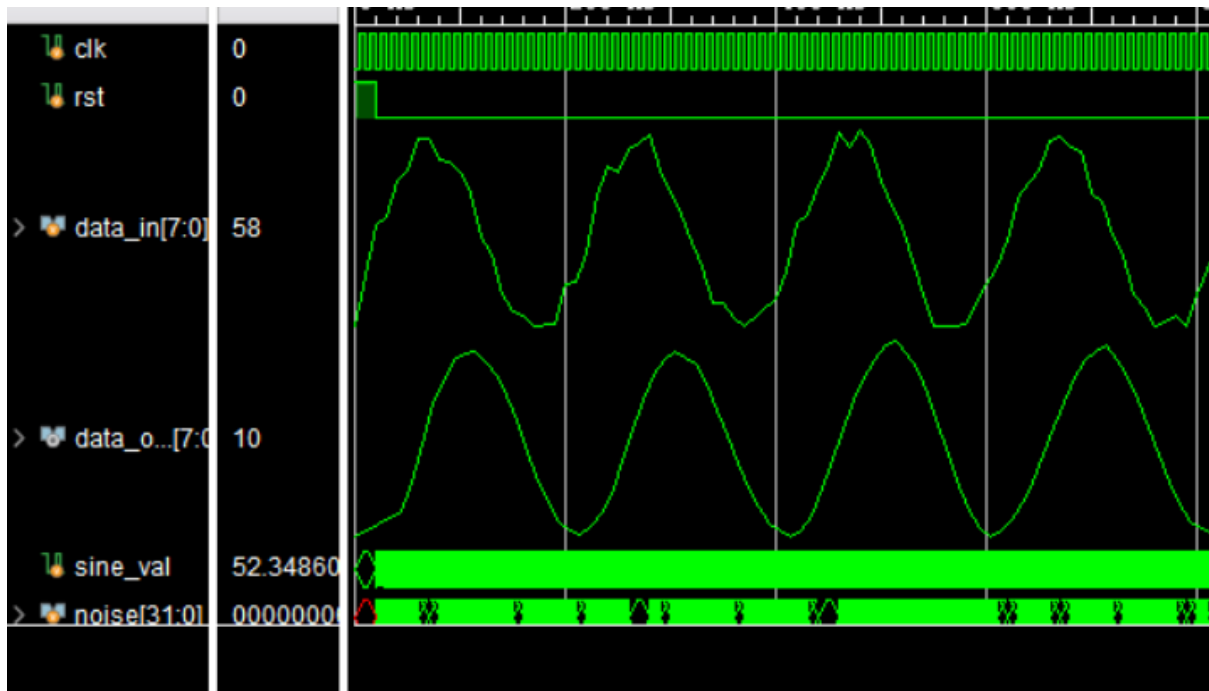
```

```

for (i = 0; i < 100; i = i + 1) begin
    sine_val = 127.0 * (1.0 +
        $sin(2.0 * 3.14159 * i / 20.0));
    noise = ($random % 21) - 10;
    temp = $rtoi(sine_val) + noise;
    if (temp > 255)
        data_in = 8'd255;
    else if (temp < 0)
        data_in = 8'd0;
    else
        data_in = temp[7:0];
    #10;
end
#100;
$stop;
end
endmodule

```

Output



Result

The moving average filter was successfully simulated using RTL Verilog in Xilinx Vivado. A sine wave corrupted with random noise was applied as the input signal through the testbench. The simulation results show that the filter effectively reduces the random noise components while preserving the sinusoidal shape of the signal.

The output waveform is smoother than the input waveform, confirming that the moving average filter operates as a low-pass FIR filter. Thus, the design and verification of the moving average filter using a noisy sine wave input were successfully completed.